

CAMP-HNSW: Hardware-Topological Co-Design for OOD-Robust ANNS via Cache Alignment and Targeted Rewiring

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ABSTRACT

Graph-based Approximate Nearest Neighbor Search (ANNS) algorithms, such as HNSW, suffer from significant performance degradation under Out-of-Distribution (OOD) queries due to local minima traps. Recent state-of-the-art (SOTA) solutions mitigate this via auxiliary structures or dynamic edge injection. However, these additive densification strategies—even when capacity-bounded—accumulate redundant edges, leading to suboptimal topological efficiency that degrades query throughput (QPS) at practical recall ranges.

We propose **CAMP-HNSW**, a Pareto-superior framework that co-designs physical data layout and topological routing to robustify ANNS indices. Our structured pipeline comprises: (1) **Cache-Aligned Memory Reordering** to physically align data and maximize L3 cache locality; (2) **Targeted Shortcut Mining** to identify critical bridging edges via historical queries; and (3) **Joint Heuristic Pruning**, which systematically evaluates each connection’s spatial necessity rather than merely filling an edge quota. By selectively replacing redundant base edges with high-survival OOD shortcuts, this joint rewiring facilitates an adaptive degree reallocation that maximizes routing efficiency.

Extensive evaluations on four datasets (ImageNet, LAION, COCO, and MainSearch) demonstrate CAMP-HNSW achieves SOTA QPS across a wide, practical recall spectrum (95%–99%). Our Joint Heuristic Pruning ensures high navigability with superior topological efficiency by dynamically adapting the local degree to the manifold complexity. Crucially, CAMP-HNSW delivers this dominant performance using a robust default configuration, providing a highly hardware-efficient routing framework for cross-modal vector search.

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The source code, data, and/or other artifacts have been made available at <https://github.com/huijunjin/CAMP-HNSW>.

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1 INTRODUCTION

As modern data ecosystems continue to scale, Approximate Nearest Neighbor Search (ANNS) has become an indispensable primitive for high-dimensional data management. It plays a performance-critical role across diverse domains, including large-scale information retrieval [1, 5, 35, 38, 41, 55], personalized recommendation engines [8, 39], and semantic question answering [29, 44]. More recently, the surge in Retrieval-Augmented Generation (RAG) within Large Language Model (LLM) workflows has further increased the demand for efficient vector indices that function as external knowledge repositories [3, 12, 28, 34, 49]. To address these multi-faceted demands, a variety of indexing paradigms have been developed, such as tree-based structures [11, 45, 46, 50, 56], quantization-based schemes [2, 4, 15–18, 25, 37, 53], hashing-based methods [9, 14, 21, 48, 52, 57], and graph-based indices [6, 13, 19, 24, 36, 40]. Among these, graph-based indices, most notably the Hierarchical Navigable Small World (HNSW) [36], represent the state-of-the-art (SOTA) by delivering an optimal trade-off between search throughput and recall for In-Distribution (ID) workloads.

Building upon our previous efforts to optimize HNSW through data-adaptive parameter adjustments [26] and proactive index repair for dynamic reliability [27], this paper investigates a critical but often overlooked dimension: robustness against Out-of-Distribution (OOD) queries. In practical deployments, such as cross-modal retrieval, the query distribution frequently deviates from the indexed base data due to an inherent modality gap [6, 31]. These OOD queries occupy different sub-manifolds, where the respective ground truths (GT) are often sparsely distributed rather than densely clustered [6, 31]. Such spatial discrepancies break the assumption of local isotropy relied upon by standard graph-based indices. This mismatch triggers *topological blindness*, where the routing algorithm becomes trapped in *OOD traps* (local minima), resulting in a sharp decline in search accuracy.

To mitigate these failures, recent robust ANNS frameworks such as RoarGraph [6] and NGFix [20] introduce auxiliary connectivity to bridge the inter-manifold gaps. While effective in restoring recall, these approaches introduce three systemic limitations: **1) Hardware-Agnostic Topology Design:** Current methods focus exclusively on graph connectivity, overlooking the physical memory layout. This hardware-agnostic design leads to substantial L3 cache misses during OOD traversal, where even a slight increase in connectivity translates into high latency overhead. **2) Structural Redundancy:** Existing solutions treat OOD shortcuts as auxiliary patches rather than integrating them into the core index. This additive approach fails to re-evaluate the utility of base edges in the presence of new shortcuts, resulting in a redundant topological structure where search paths are cluttered with spatially

redundant connections. 3) **Fragility to Dataset-Specific Tuning:** The effectiveness of these indices often depends on rigid configurations—such as fixed projection depths or specific hardness detection thresholds—that are highly sensitive to the underlying data manifold. Such sensitivity hinders a seamless and robust deployment across diverse, real-world database environments.

In this paper, we argue that OOD robustness is best achieved through a holistic co-design of the physical data layout and topological routing. We propose **CAMP-HNSW** (Cache-Aligned Mining and Pruning), a Pareto-superior framework that preserves the native HNSW greedy routing engine while systematically eliminating its hardware and topological bottlenecks. Our approach addresses the aforementioned limitations as follows: First, to resolve hardware-agnostic designs, we integrate a **Cache-Aligned Memory Reordering** module. By utilizing MiniBatchKMeans [43] to physically group topologically close nodes, we minimize L3 cache misses. Second, to eliminate structural redundancy, we combine **Targeted Shortcut Mining with Joint Heuristic Pruning**. This synergistic process proactively identifies vital OOD shortcuts while forcing them to compete for survival against existing edges, ensuring maximum navigability within a bounded degree. Finally, the overall design facilitates deployability through a robust default configuration, overcoming the fragility to parameter tuning inherent in existing solutions.

Our contributions are summarized as follows:

- We demonstrate that cache-aligned physical data reordering, when integrated with graph-based indexing, significantly boosts hardware efficiency by translating topological proximity into physical spatial locality.
- We propose a targeted topological rewiring mechanism that adaptively refines the graph structure by mining OOD shortcuts and subjecting them to a joint competitive pruning process, ensuring high navigability while maintaining a strictly controlled out-degree.
- Through multi-objective optimization, we establish a deterministic, robust default parameter set. This robust default configuration proves the architectural stability of CAMP-HNSW, eliminating the heavy reliance on exhaustive per-dataset hyperparameter tuning.
- Through extensive empirical evaluations across four diverse datasets, we demonstrate that CAMP-HNSW achieves SOTA query throughput at practical, high-recall regimes (95%–99%) while preserving superior build efficiency and strictly bounded memory limits.

2 BACKGROUND & MOTIVATION

2.1 The Anatomy of OOD Traps and Topological Blindness

Graph-based ANNS algorithms, such as HNSW [36], operate on the fundamental assumption of local isotropy, where queries and dataset vectors are drawn from the same underlying manifold (ID). Under this assumption, greedy routing effectively navigates toward the target by minimizing the distance to the query at each hop, relying on the cluster property where nearest neighbors are spatially concentrated.

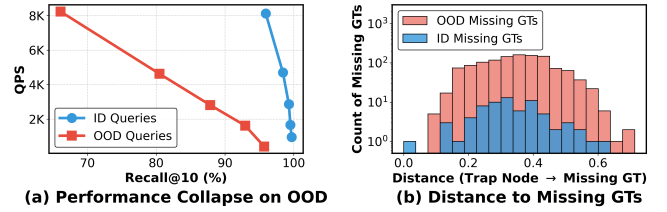


Figure 1: Motivation of OOD routing failures on the LAION dataset. The base HNSW suffers from (a) substantial throughput degradation, which is primarily caused by (b) topological blindness where missing GTs are physically distant from local minima (traps).

However, in modern multi-modal applications, OOD queries exhibit pronounced spatial deviations that break these core assumptions [6]. Specifically, the inherent modality gap (e.g., text vs. image) forces OOD queries to occupy disjoint sub-manifolds located far from the support of the base data distribution. Furthermore, unlike ID workloads where a query’s k -nearest neighbors (k -NNs) are tightly clustered, the true targets of an OOD query are frequently highly fragmented and distant from one another in the embedding space.

These properties cause a sharp performance decline in standard indices. As illustrated in Figure 1(a), using the exact same base HNSW index, OOD queries suffer from substantial QPS degradation compared to ID queries to achieve comparable recall. The greedy search fails because HNSW’s local connectivity is optimized for navigating within a single, continuous manifold, making it incapable of bridging across the gaps between the query and its sparse, scattered neighbors.

To analyze this failure, we define a *trap node* as the final node where greedy search halts due to a local minimum—where no structural neighbors are closer to the query than the node itself. Figure 1(b) reveals a counter-intuitive phenomenon: the routing failure is not primarily caused by vast physical distances across the entire embedding space. A large fraction of OOD queries fail even when the missing GT is in immediate physical proximity to the trap node. We term this phenomenon as *topological blindness*. The algorithm approaches the target’s physical vicinity but remains structurally disconnected because the base graph lacks the explicit edges required to cross the inter-manifold gap, effectively creating OOD traps.

2.2 The Limitations of Additive Densification

The observation of topological blindness exposes the inherent limitations of recent methods that rely on additive densification. While effective at bridging distribution gaps, these methodologies often introduce structural rigidity or suboptimal efficiency. For instance, RoarGraph [6] constructs an auxiliary bipartite projection graph to connect manifolds. However, this approach requires a substantial “warm-up” period with a large historical query set and imposes construction complexity, as the bipartite structure must be meticulously maintained to avoid structural staleness.

Similarly, NGFix [20] dynamically identifies hard queries and injects routing edges to heal disconnected components. Although

NGFix enforces a degree bound via a capacity parameter, it operates as an additive “detect-and-fix” patch rather than a holistic optimization. Because it lacks a global replacement mechanism that re-evaluates existing base edges alongside new shortcuts, the resulting topology often suffers from redundant connectivity. This additive approach ignores the hardware-level impact; evaluating these redundant neighbors across a physically scattered memory layout disrupts spatial locality and causes frequent L3 cache thrashing. Therefore, we argue that true OOD robustness requires a holistic co-design of physical spatial locality and unified degree management, which we realize through the CAMP-HNSW framework.

3 METHODOLOGY

In this section, we detail the architecture of CAMP-HNSW. Guided by the principle of hardware-algorithm co-design, our framework operates as a streamlined, highly parallelized pipeline (illustrated in Figures 2 and 3). Rather than inventing overly complex mathematical structures, we systematically align the physical memory layout and topologically rewire the base HNSW graph to resolve the OOD trapping problem without excessive memory bloat.

3.1 Preliminaries and Problem Formulation

Greedy Routing and Local Minima. Let $X = \{x_1, \dots, x_N\} \subset \mathbb{R}^D$ be the base dataset consisting of N vectors in D -dimensional space. We define $G_{base} = (V, E_{base})$ as the proximity graph constructed over X , where V represents the set of nodes corresponding to these vectors and E_{base} denotes the edges. During standard greedy routing for a query q , the algorithm navigates progressively towards the closest node. The routing terminates at a local minimum $v_t \in V$ if no neighbor of v_t is closer to q than v_t itself:

$$\text{dist}(q, v_t) \leq \text{dist}(q, u), \quad \forall u \in N_{base}(v_t) \quad (1)$$

where $N_{base}(v_t) = \{u \mid (v_t, u) \in E_{base}\}$.

The Monotonic Search Network (MSN) Property. A proximity graph successfully guarantees exact routing if it satisfies the MSN property [13]. For any query q and starting node p , the MSN property dictates that there must exist at least one connected path $\{p, v_1, v_2, \dots, v^*\}$ to the GT v^* such that the distance to q strictly decreases at each hop (i.e., $\text{dist}(q, v_i) < \text{dist}(q, v_{i-1})$).

The OOD Trap and Topological Blindness. We assume the base dataset X is drawn from an ID P_{ID} . Under standard scenarios ($q \sim P_{ID}$), the MSN property largely holds, and v_t is typically the global nearest neighbor. However, in modern multi-modal applications, queries are frequently drawn from an OOD (P_{OOD}), inherently introducing a cross-modal modality gap (e.g., text queries searching image vectors). This topological mismatch significantly distorts the spatial landscape. We formally define an *OOD trap* as the topological state where the routing halts at v_t , yet a true GT v^* exists such that:

$$\text{dist}(q, v^*) < \text{dist}(q, v_t) \quad \text{and} \quad v^* \notin N_{base}(v_t) \quad (2)$$

In this state, to reach the true target v^* from the trap v_t , the routing algorithm must temporarily traverse through nodes that are further from q than v_t is. This requirement to move “away” from the query explicitly breaks the greedy MSN property, causing the algorithm to remain topologically blind to the target. Following standard practices for cross-modal indexing, we utilize a historical

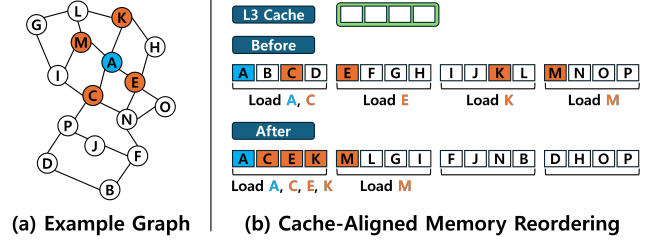


Figure 2: Illustration of Cache-Aligned Memory Reordering (Phase 1): (a) Logical proximity graph, and (b) Comparison of physical memory layouts before and after reordering.

query set $Q_{cal} \sim P_{OOD}$ to explicitly capture and bridge these structural gaps. Generalization is then evaluated on a strictly disjoint unseen set $Q_{test} \sim P_{OOD}$. Our objective is to design a refined graph $G_{CAMP} = (V, E_{CAMP})$ and a hardware-efficient memory layout X' that introduces OOD shortcuts (v_t, v^*) bounded by a maximum degree M , effectively restoring monotonic routability across modality gaps without substantial memory bloat or cache thrashing.

Base Graph Architecture. Following recent advancements in OOD-robust graph indexing [20, 36], we construct G_{base} exclusively using the bottom layer (Level 0) of HNSW, where the majority of local minima traps occur. Furthermore, to prevent entry point (EP) bias—which can be exacerbated by our physical memory chunking—we strictly fix the global EP to the real data vector closest to the geometric centroid of the dataset. This ensures that the greedy routing always initiates from the macroscopic center of the distribution, providing a stable, unbiased baseline for our subsequent topological rewiring.

3.2 System Overview

To bridge the fundamental gap between hardware efficiency and topological robustness, CAMP-HNSW is architected as a streamlined, three-phase pipeline. It fundamentally operates on top of a hardware-optimized memory foundation (Figure 2), which subsequently enables our targeted topological rewiring mechanism (Figure 3).

Phase 1: Cache-Aligned Memory Reordering. Before altering the graph topology, we first resolve the inherent memory access bottleneck of HNSW. As shown in Figure 2, by physically grouping spatially adjacent vectors into contiguous memory blocks via lightweight clustering, we ensure that standard local neighborhood traversals remain safely confined within the CPU’s L3 cache hierarchy.

Phase 2: Targeted Shortcut Mining. With the cache-aligned base graph anchored, the system identifies exact topological blind spots. As shown in Figure 3(a), standard greedy routing for an OOD query q often halts prematurely at an OOD trap (e.g., local minima nodes A , B , and C), failing to reach the true targets (D and E). To explicitly repair this broken monotonic search property, the system generates raw bidirectional shortcut candidates connecting the trapped neighborhood to the missing targets, as depicted by the dashed lines in Figure 3(b).

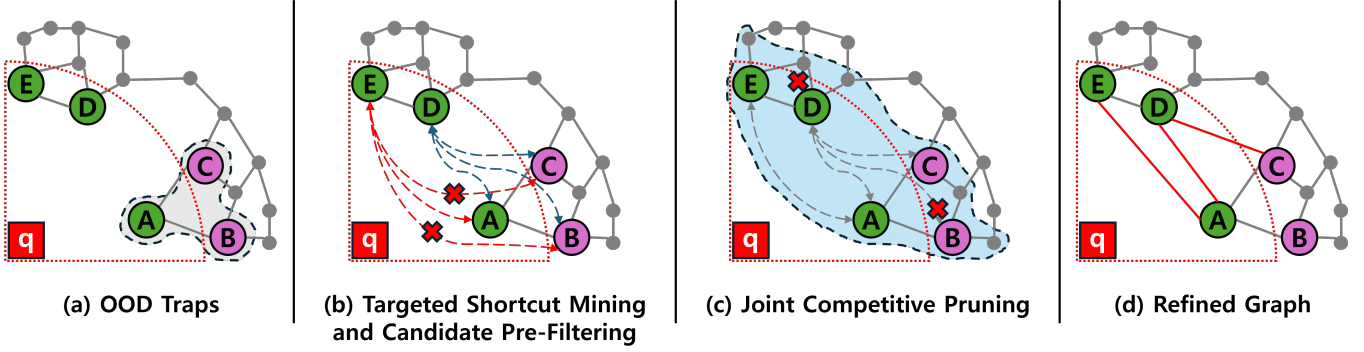


Figure 3: The core algorithmic pipeline of CAMP-HNSW, illustrating Targeted Shortcut Mining (Phase 2) and Two-Stage Joint Heuristic Pruning (Phase 3).

Phase 3: Joint Heuristic Pruning. Injecting all raw candidate edges would cause severe memory bloat and degrade routing throughput. Thus, the system employs a two-stage pruning mechanism. First, as illustrated by the red cross marks in Figure 3(b), it strictly filters out spatially redundant candidate edges. Next, as shown in Figure 3(c), the surviving shortcuts and the original base edges undergo a joint competition bounded by the maximum out-degree limit M (e.g., $M = 4$ in our illustration). Figure 3(d) displays the refined topology: redundant base edges are actively evicted, and the newly established global bridges (highlighted in red) successfully restore OOD navigability without expanding the overall memory footprint.

Inherited vs. Novel Components. While adopting historical calibration queries and ground truths from prior frameworks (e.g., RoarGraph, NGFix) and retaining the classic HNSW heuristic, our primary novelty is the joint replacement-based pruning (Phase 3). Unlike purely additive methods, CAMP-HNSW forces base edges and OOD shortcuts to compete, preventing degree bloat while optimizing cross-manifold navigability.

3.3 Phase 1: Cache-Aligned Memory Reordering

Graph-based indices inherently suffer from a critical memory access bottleneck. Traversing from a node to its topological neighbors typically results in random memory accesses. As illustrated in the proximity graph of Figure 2(a), to evaluate the neighbors of node A , the system must fetch vectors C , E , K , and M . However, under a standard sequential insertion layout, these topologically adjacent nodes are scattered arbitrarily across the physical RAM. Because raw high-dimensional datasets are rarely pre-sorted by spatial proximity, this scattered access pattern becomes the operational norm. As demonstrated in Figure 2(b, Before), gathering these five nodes triggers scattered reads across fragmented memory pages and non-adjacent cache lines, wasting memory bandwidth and causing substantial CPU stalls.

To preemptively eliminate this hardware bottleneck, CAMP-HNSW physically aligns the memory layout with the expected topological structure. We formulate this as an optimized data re-ordering problem utilizing MiniBatchKMeans.

Rationale for MiniBatchKMeans. Our objective is not to achieve absolute clustering convergence, which incurs substantial

$O(I \cdot K \cdot N \cdot D)$ overhead. Here, I denotes the number of iterations and K represents the target cluster count. Instead, our task strictly requires a fast, low-overhead approximation that macroscopically groups spatially adjacent vectors to maximize cache hit rates. By utilizing the MiniBatch variant, we significantly reduce the offline computational penalty while delivering highly effective spatial chunking.

Algorithm Execution and Complexity. Algorithm 1 details this pipeline. To accelerate preprocessing, we uniformly sample merely a fraction of the dataset (ρ) to train the cluster centroids (Lines 1-3), taking $O(\rho N \cdot K \cdot D)$ time. We then assign all N vectors to their nearest centroid (Line 4), consuming $O(N \cdot K \cdot D)$. During the packing phase (Lines 6-12), the algorithm iterates through each cluster, physically appending vectors sequentially into a contiguous array X' and constructing bidirectional mapping tables. This sorting and repacking strictly require $O(N)$ time and $O(N)$ auxiliary space. Given that $K \ll N$ and ρ is small, this phase adds negligible preprocessing overhead.

Hardware Impact and Mapping Overhead. The physical chunking granularity is controlled by the cluster count K . To maximize cache residency and minimize random DRAM access penalties, the system dynamically determines K such that an average manifold cluster can be staged entirely within the CPU’s active L3 cache hierarchy. Assuming standard 32-bit floating-point vectors, the theoretical lower bound for K is mathematically formulated as a hardware constraint:

$$K \geq \frac{N \times D \times 4 \text{ bytes}}{C_{L3}}$$

where C_{L3} is the available L3 cache capacity. By aligning K with this hardware constraint, fetching a node allows the hardware prefetcher to seamlessly load its topological neighbors from contiguous memory regions into adjacent cache lines. While K-Means imbalance and cache associativity prevent absolute residency guarantees, this hardware-driven boundary provides a highly effective, tuning-free baseline.

The practical hardware impact is visualized in Figure 2(b, After). Because node A and its neighbors C , E , and K share the same local centroid, Algorithm 1 packs them into contiguous memory blocks. Consequently, fetching node A empowers the CPU to preload C , E ,

Algorithm 1: Cache-Aligned Memory Reordering

Input: Raw Dataset $X \in \mathbb{R}^{N \times D}$, Clusters K , Sample Ratio ρ
Output: Reordered Dataset X' , Mapping Tables

$\mathcal{M}_{new}, \mathcal{M}_{old}$

```
1  $N_{sample} \leftarrow \lfloor N \times \rho \rfloor$ ;  
2  $X_{sample} \leftarrow \text{UniformSample}(X, N_{sample})$ ;  
3  $C \leftarrow \text{MiniBatchKMeans}(X_{sample}, K)$ ;  $\triangleright$  Train Centroids  
4  $L \leftarrow \text{PredictLabels}(X, C)$ ;  $\triangleright$  Assign all  $x_i \in X$   
5  $X' \leftarrow \emptyset, idx_{new} \leftarrow 0$ ;  
6 for  $k \leftarrow 0$  to  $K - 1$  do  
7    $V_k \leftarrow \{i \mid L[i] == k\}$ ;  
8   for  $i \in V_k$  do  
9     Append  $X[i]$  to  $X'$ ;  $\triangleright$  Physically pack memory  
10     $\mathcal{M}_{new}[i] \leftarrow idx_{new}$ ;  $\triangleright$  Mapping: Old  $\rightarrow$  New  
11     $\mathcal{M}_{old}[idx_{new}] \leftarrow i$ ;  $\triangleright$  Mapping: New  $\rightarrow$  Old  
12     $idx_{new} \leftarrow idx_{new} + 1$ ;  
13 return  $X', \mathcal{M}_{new}, \mathcal{M}_{old}$ ;
```

and K into adjacent cache lines or the same cache-resident memory block via spatial locality, significantly reducing Translation Lookaside Buffer misses and scattered L3 cache loads. Finally, the $O(1)$ mapping tables ($\mathcal{M}_{new}, \mathcal{M}_{old}$) consume merely ~ 8 MB for a 1-million scale dataset. Crucially, the logical graph is constructed directly using these new contiguous physical indices (idx_{new}). As a result, the routing phase completely bypasses ID translation and operates at native memory speed. The translation back to the original dataset IDs via $\mathcal{M}_{old}$ occurs strictly once at the end of the query to formulate the final Top-K results, guaranteeing zero latency penalty during the search phase.

While cache-aligned memory reordering is a fundamental systems optimization that generally benefits standard routing, its integration is structurally essential for our specific OOD pipeline. Because OOD shortcuts inherently force the router to make distant memory leaps across the manifold, they severely degrade L3 cache hit rates. Phase 1 acts as a foundational hardware co-design to absorb these irregular memory access patterns induced by the subsequent OOD edge injections.

3.4 Phase 2: Targeted Shortcut Mining

With the physically aligned dataset X' loaded and the base graph G_{base} anchored at the global centroid, the system begins identifying the exact topological blind spots. To accurately model these OOD traps without introducing artificial bias, we utilize a calibration query set, Q_{cal} , sampled directly from the historical OOD workload distribution.

Identifying Topological Traps (Algorithm 2, Lines 2-5). The mining process begins by diagnosing where the standard routing algorithm fails. For each calibration query $q \in Q_{cal}$, the system performs an exploratory greedy search parameterized by a search depth variable, ef_{mine} , to retrieve a terminal local neighborhood, F_q . This parameter strictly governs the intensity of the exploration. A smaller ef_{mine} ensures rapid offline preprocessing but risks terminating too early, yielding an overly narrow F_q that fails to capture

Algorithm 2: Targeted Mining and Edge Rewiring

Input: Base Graph G_{base} , Calibration Queries Q_{cal} ,
Pre-computed GTs T , ef_{mine} , N_{mine} , Max Degree M

Output: Refined Graph G_{CAMP}

$\triangleright$ **Phase 2: Targeted Shortcut Mining**

```
1  $E_{cand} \leftarrow \emptyset$ ;  
2 foreach  $q \in Q_{cal}$  in parallel do  
3    $F_q \leftarrow \text{Search}(G_{base}, q, ef_{mine})$ ;  $\triangleright$  Terminal  
   neighborhood  
4    $T_q \leftarrow \text{PrecomputedGT}(q, T)$ ;  $\triangleright O(1)$  offline label  
   lookup  
5    $M_q \leftarrow T_q \setminus F_q$ ;  $\triangleright$  Identify topological blind  
   spots  
6   if  $M_q \neq \emptyset$  then  
7      $\hat{F}_q \leftarrow \text{ExtractTop}(F_q, q, N_{mine})$ ;  $\triangleright$  Prioritize by  
     spatial proximity  
8      $\hat{M}_q \leftarrow \text{ExtractTop}(M_q, q, N_{mine})$ ;  
9     foreach  $u \in \hat{F}_q, v \in \hat{M}_q$  do  
10    | ThreadLocalAdd( $E_{cand}, (u, v)$ );
```

$\triangleright$ **Phase 3: Joint Heuristic Pruning**

```
11 foreach  $v \in V$  in parallel do  
12    $N_{cand}(v) \leftarrow \{u \mid (v, u) \in E_{cand}\}$ ;  $\triangleright$  Extract  
   candidate neighbors for  $v$   
13    $N_{cand}(v) \leftarrow \text{HeuristicSelection}(N_{cand}(v), \infty)$ ;  $\triangleright$  1.  
   Candidate Pre-Filtering  
14    $P(v) \leftarrow N_{base}(v) \cup N_{cand}(v)$ ;  $\triangleright$  Joint Edge Pool  
15    $N_{final}(v) \leftarrow \text{HeuristicSelection}(P(v), M)$ ;  $\triangleright$  2. Joint  
   Competitive Pruning  
16   Lock( $v.mutex$ );  $\triangleright$  Fine-Grained Node-Level  
   Locking  
17   Update  $G_{CAMP}$  edges for  $v$  with  $N_{final}(v)$ ;  
18   Unlock( $v.mutex$ );  
19 return  $G_{CAMP}$ ;
```

the true boundaries of the trap. Conversely, a larger ef_{mine} comprehensively maps the trapped neighborhood at the expense of increased offline traversal time.

Crucially, F_q represents the local minimum where the greedy search halts. While F_q may successfully capture a fraction of the exact GTs, the remaining non-GT nodes in F_q act as a collective topological trap. Simultaneously, the system retrieves the exact GT set T_q . By computing the set difference, $M_q = T_q \setminus F_q$, we isolate the missing GTs that are topologically unreachable from the local minimum, regardless of their absolute physical distance in the embedding space. The exact ground truth sets (T_q) are pre-computed offline, a standard prerequisite in query-aware graph methodologies. Consequently, our reported build times strictly reflect the actual algorithmic overhead: base graph construction, local minima traversals, edge generation, and structural pruning.

Cross-Manifold Bridge Generation (Algorithm 2, Lines 6-8). If M_q is not empty, the algorithm must explicitly link the

OOD traps to the missing GTs. However, fully connecting every local minimum node in F_q to every missing target in M_q via a Cartesian product would trigger a combinatorial memory explosion. To prevent this, the maximum number of explicitly connected nodes is strictly regulated by a budget parameter, N_{mine} . Specifically, the system extracts bounded subsets $\hat{F}_q \subseteq F_q$ and $\hat{M}_q \subseteq M_q$ by retaining only the top- N_{mine} nodes from each set, prioritized strictly by their spatial proximity to the query q , before generating the bidirectional candidate bridges:

$$E_{cand}(q) = \{(u, v), (v, u) \mid u \in \hat{F}_q, v \in \hat{M}_q\} \quad (3)$$

where $|\hat{F}_q| \leq N_{mine}$ and $|\hat{M}_q| \leq N_{mine}$.

The N_{mine} parameter dictates the fundamental system trade-off between structural robustness and offline complexity. A conservative N_{mine} keeps the offline mining exceptionally fast but may fail to supply sufficient structural bridges to escape OOD gaps. Conversely, an aggressively large N_{mine} triggers a quadratic explosion in candidate generation—strictly bounded by $O(N_{mine}^2)$ per query—and heavily overburdens the subsequent Phase 3 pruning stage, yielding diminishing returns in actual online recall.

Because these explicit cross-manifold connections directly overcome the topological blindness, they tentatively restore the monotonic search property required to reach M_q . We detail the empirical optimization of these parameters in Section 4.6.

System-Level Parallelism. Applying this generation process across the entire calibration set yields millions of candidate edges. To efficiently process this large volume without bottlenecking the system, the intensive graph traversal and edge generation are entirely offloaded to a custom C++ core engine. We adopt robust concurrent engineering practices to maximize multi-core utilization. First, we leverage OpenMP with a `schedule(dynamic, 64)` directive. This dynamically assigns small chunks of 64 queries to idle physical cores, ensuring optimal load balancing. Second, to avoid global mutex contention when appending newly discovered edges, we utilize a Thread-Local Storage mechanism. Each thread populates its own private candidate buffer (as invoked by `ThreadLocalAdd`), merging them sequentially only at the final step. This lock-free pipeline ensures the mining phase operates in strictly bounded $O(|Q_{cal}| \cdot N_{mine}^2 / C)$ wall-clock time across C physical cores, demonstrating high architectural scalability.

3.5 Phase 3: Joint Heuristic Pruning

A naive injection of all mined candidate edges (E_{cand}) into the base graph would trigger substantial memory bloat and degrade routing throughput. CAMP-HNSW counters this by implementing a two-stage dynamic structural rewiring mechanism.

Stage 1: Candidate Pre-Filtering (Algorithm 2, Lines 9-11). The raw E_{cand} pool contains many dense, spatially redundant connections. Before modifying the base graph, the system processes each node $v \in V$ in parallel, extracting its specific candidates $N_{cand}(v)$ and subjecting them to the native Relative Neighborhood Graph (RNG) heuristic of HNSW. Rather than introducing a complex custom mathematical filter—which risks distorting the spatial manifold and disrupting the greedy routing—we intentionally re-purpose the standard RNG logic. By evaluating our explicitly mined cross-manifold bridges through this native heuristic, we ensure the

surviving shortcuts are inherently compatible with the base routing engine.

Specifically, the internal `HeuristicSelection` evaluates candidate targets for node v sequentially in order of increasing distance. A candidate edge (v, u) is retained only if it guarantees spatial diversity—meaning u is strictly closer to v than to any previously accepted neighbor w :

$$\text{dist}(v, u) < \text{dist}(u, w), \quad \text{for all previously accepted } w \quad (4)$$

This preliminary filtering stage effectively shrinks the $N_{cand}(v)$ pool by discarding redundant parallel pathways, leaving only the most vital structural bridges intact before the actual graph rewiring begins.

Stage 2: Joint Competitive Pruning (Algorithm 2, Lines 12-13). For a given node $v \in V$, the system combines its original base edges $N_{base}(v)$ and the surviving OOD shortcuts $N_{cand}(v)$ into a single joint edge pool $P(v)$. The algorithm forces this combined pool to undergo the RNG heuristic selection again, strictly bounded by the maximum out-degree parameter M . In this competitive environment, both redundant base edges and suboptimal shortcuts are actively pruned to adhere to the fundamental memory limits.

Fine-Grained Locking and Complexity (Algorithm 2, Lines 14-16). Following standard practices for concurrent data structures, modifying edges across millions of vectors is handled via fine-grained node-level locks (e.g., utilizing `std::mutex` per node). As shown in Lines 14-16, locking only the specific adjacency list of a target node allows dozens of threads to prune and rewire independent regions of the graph simultaneously with minimal thread contention.

This joint heuristic pruning phase is highly efficient. For each node v , the joint selection requires $O(|P(v)| \log |P(v)|)$ time for initial distance sorting, followed by $O(|P(v)| \cdot M \cdot D)$ time for the spatial diversity checks. Because Phase 1 guarantees that the majority of the surviving local proximity edges are cache-aligned, the extensive distance computations during this rewiring phase—and subsequent online routing—remain highly hardware-efficient.

4 EXPERIMENTAL EVALUATION

In this section, we evaluate CAMP-HNSW against SOTA baselines to answer four core research questions (RQs). **(1) RQ1 (Online Performance):** Does CAMP-HNSW deliver superior QPS across practical recall targets on diverse OOD manifolds? (Section 4.2). **(2) RQ2 (System Efficiency):** How does our co-design impact the offline memory footprint, build time, and L3 cache efficiency? (Sections 4.3–4.4). **(3) RQ3 (Architecture & Stability):** Are all proposed components structurally necessary, and does the framework operate robustly default parameters? (Sections 4.5–4.6). **(4) RQ4 (Extreme Robustness):** How does the system perform under strict operational constraints, such as zero-penalty ID routing and sparse-query cold-start scenarios? (Section 4.7).

4.1 Experimental Setup

Hardware Environment and Implementation. All methods are implemented in C++ (GCC -O3) and executed on an Intel Xeon w5-2465X (32-core, 33.75MB L3, 125GB RAM). We allocate 32 threads

Table 1: Summary of Evaluated Benchmark Datasets. (N : Size, D : Dimension, Q_C : Calibration Queries, Q_T : Test Queries)

Dataset	N	D	$ Q_C $	$ Q_T $	LID_D	LID_Q	Shift	Dist.
<i>OOD Datasets</i>								
ImageNet	1.3M	640	315K	1K	23.5	203.4	2.2	IP
LAION	1.0M	512	1.0M	1K	36.3	157.4	2.0	IP
COCO	282K	768	114K	1K	19.2	718.4	4.8	IP
MainSearch	11.2M	256	1.0M	10K	22.6	127.0	4.2	IP
<i>ID Datasets</i>								
GloVe	992K	200	200K	1K	49.9	50.8	1.0	Cos
ImageNet	1.0M	512	200K	1K	25.0	25.4	1.0	IP

for offline construction and strictly restrict online search to a single thread.

Datasets. We evaluate CAMP-HNSW on four large-scale OOD benchmarks featuring severe cross-modal fragmentation. To ensure rigorous evaluation under established modality gaps, we utilize the multimodal (text-to-image) embeddings and query splits provided by the VIBE benchmark [22] for three canonical datasets: ImageNet [10] (a densely clustered image manifold), LAION [42] (a highly sparse and fragmented manifold), and COCO [32] (a complex multi-object distribution). Additionally, to stress-test architectural scalability, we include MainSearch [20], a large-scale industrial dataset.

We utilize their standard test queries (Q_{test}) for evaluation and calibration sets (Q_{cal}) for shortcut mining. Furthermore, to verify that our structural optimizations do not compromise ID efficiency, we also evaluate the framework on GloVe (dense word embeddings) and an ID-variant of ImageNet. Since these ID datasets lack designated calibration queries, we partitioned 200,000 base vectors to serve as Q_{cal} , reserving the remainder for the foundational base index. Detailed dataset statistics and their evaluated distance metrics are summarized in Table 1. For all query-aware methods evaluated, the exact ground truth computation for calibration queries is executed strictly as an offline preprocessing step and is completely excluded from the reported index build times.

To formalize the structural hardness of these datasets, Table 1 reports the Local Intrinsic Dimensionality (LID) and OOD Shift Ratio (Shift). Specifically, the Base LID indicates intrinsic uniformity, where higher values (e.g., LAION) require more aggressive edge pruning. The Query LID reflects the target distribution’s complexity, while the OOD Shift Ratio measures the cross-modal gap. Consequently, datasets combining high internal uniformity, complex query spaces, and severe modality gaps inherently incur higher indexing overhead.

Baselines and Parameter Configurations. CAMP-HNSW is evaluated against HNSW (the base ID graph), RoarGraph (cross-modal bipartite projection), and NGFix (dense edge injection). To ensure a fair comparison, the fundamental parameters governing the base proximity topology—specifically the maximum out-degree ($M = 32$) and the construction search depth ($ef_{construction} = 200$)—are configured identically across all evaluated methods, including their respective structural equivalents in RoarGraph and NGFix. For the baseline OOD adaptations, we strictly adhere to

the optimal hyperparameters recommended by their respective authors to ensure maximum baseline performance. For CAMP-HNSW, rather than performing exhaustive per-dataset tuning, we deploy a single, robust default configuration derived from static hardware constraints and empirical parameter tuning on 1M-scale reference datasets (Section 4.6). We apply this exact default configuration uniformly across all evaluated datasets and scales, demonstrating the architectural stability and practical usability of our framework.

4.2 Online Search Performance

We evaluate the online search performance of CAMP-HNSW against baseline algorithms. To provide a detailed analysis of system robustness, Figure 4 visualizes the Recall@ K versus QPS Pareto frontiers across four datasets for search targets $K = 1, 10$, and 100. The results reveal clear operational trade-offs among the baselines and demonstrate the architectural advantage of CAMP-HNSW across different search granularities.

Consistent Pareto Superiority at Extremes ($K \in \{1, 100\}$). When the search task requires strict precision (Recall@1) or deep exploration (Recall@100), CAMP-HNSW achieves clear Pareto dominance. At Recall@1, standard methods struggle to pinpoint the exact optimal node under severe modality gaps. CAMP-HNSW’s explicit short-circuiting of OOD traps ensures direct paths to the GT, resulting in a distinct vertical gap in QPS. Similarly, at Recall@100, where the algorithm must evaluate a massive pool of neighbors, the advantage of our co-design shines. As the search queue expands, methods relying on additive edge injection (e.g., NGFix) suffer from compounded distance computation overheads. CAMP-HNSW curbs this overhead because our Joint Heuristic Pruning enforces a strict degree bound, ensuring that these extensive neighbor evaluations remain highly cache-efficient, thereby yielding strictly superior throughput.

Operational Versatility vs. Baseline Limitations. A comprehensive analysis across the practical $K = 10$ setting clearly exposes the structural bottlenecks of the baselines. Base HNSW consistently fails to maintain practical throughput, illustrating the severe impact of topological blindness. While RoarGraph covers a broad recall spectrum and achieves high QPS at the lower precision end, its throughput degrades significantly faster than other robust methods as the recall target increases. Consequently, at identical, high-recall targets, RoarGraph’s QPS is strictly sub-optimal compared to NGFix and CAMP-HNSW. NGFix, conversely, shows resilience on specific distributions such as ImageNet and COCO, but suffers from a rigid “throughput ceiling.” Because it relies on injecting dense edges, its Pareto curve is remarkably steep—securing high throughput only at the extreme recall tail ($> 98\%$) but failing to provide optimal QPS across a wider, flexible spectrum.

These individual limitations are most vividly exposed on the MainSearch dataset. Here, the operational curves exhibit almost zero overlap, establishing a definitive and unambiguous performance hierarchy: CAMP-HNSW strictly dominates across all recall targets, followed by NGFix, RoarGraph, and finally the collapsed base HNSW.

In contrast to the baselines, the CAMP-HNSW successfully bypasses all these structural limitations. Our system maintains the

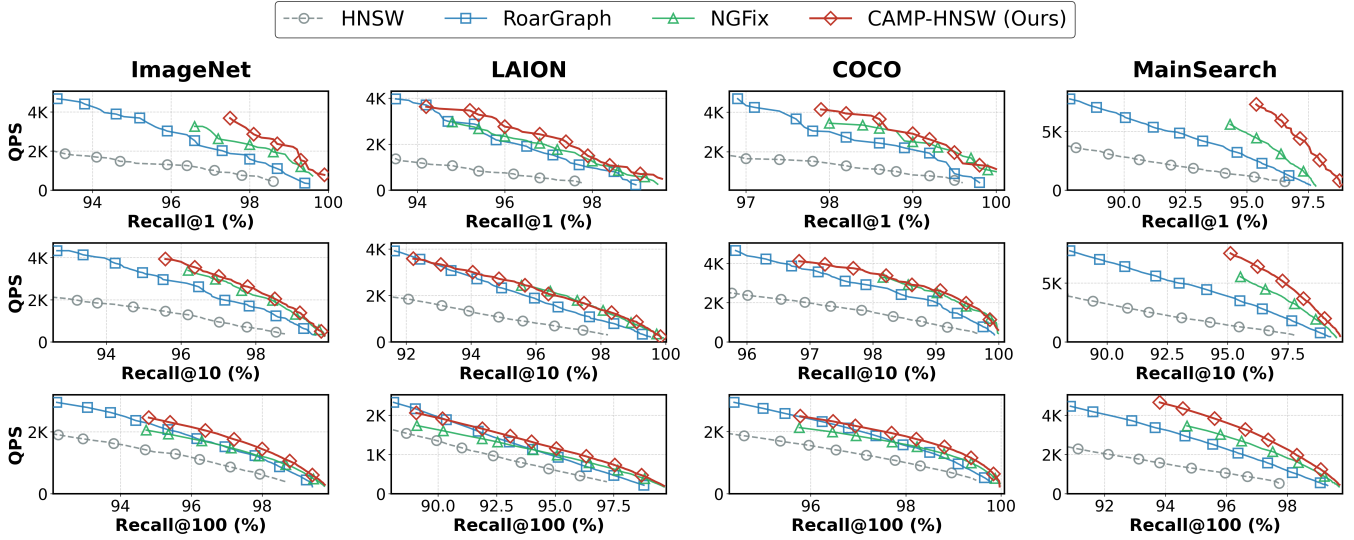


Figure 4: Performance comparison of CAMP-HNSW and baseline algorithms on four diverse datasets ($K \in \{1, 10, 100\}$).

highest QPS across a broad spectrum of practical recall targets, regardless of the underlying manifold complexity or dataset scale. We attribute this consistent performance to the synergistic co-design of physical memory alignment and topological rewiring, the individual impacts of which are systematically dissected in the following sections.

4.3 Topological Efficiency and Offline Costs

Achieving high online throughput typically incurs significant offline costs in memory footprint and index construction time. To demonstrate that CAMP-HNSW avoids these systemic penalties, we evaluate the internal graph rewiring dynamics and their direct impact on the final memory footprint and computational overhead.

Adaptive Topological Rewiring Dynamics. Table 2 details the internal edge composition before and after Joint Heuristic Pruning. The composition shift reveals a targeted mining process: over 98.6% of candidate shortcuts survive the joint competition ($\leq 1.4\%$ drop ratio). This confirms that our heuristic selectively mines topologically critical edges rather than inflating the graph with low-value candidates.

To accommodate these essential structural additions without inflating the overall memory footprint, CAMP-HNSW aggressively identifies and prunes redundant local connections within the base HNSW graph. Crucially, this deletion is intrinsically manifold-adaptive. Under challenging OOD conditions like the LAION dataset, the standard base topology proves insufficient; our heuristic actively prunes 42.1% (roughly 9.6 million edges) of the base HNSW topology, effectively replacing the inefficient paths with a globally connected routing network. Conversely, on heavily clustered data like ImageNet and MainSearch, the dense local proximity graph already captures many necessary paths. Consequently, the base edge deletion remains highly conservative (6.79% and 1.54%, respectively), allowing the algorithm to preserve the reliable local

Table 2: Graph Edge Composition Before and After Joint Heuristic Pruning. Percentages indicate the reduction rate.

Metric	ImageNet	LAION	COCO	MainSearch
Base HNSW	26.1M	22.9M	5.1M	192.7M
Mined Shortcuts	12.5M	24.9M	2.3M	47.6M
Refined HNSW	24.3M (-6.8%)	13.3M (-42.1%)	4.8M (-5.0%)	189.7M (-1.5%)
OOD Shortcuts	12.5M (-0.3%)	24.5M (-1.4%)	2.3M (-0.3%)	47.2M (-0.7%)

Table 3: In-Depth Shortcut Utility and Trap Escape Analysis.

Dataset	Access Freq.	Trapped	Escaped	Escape Rate
ImageNet	31.7%	80	76	95.0%
LAION	44.4%	78	64	82.1%
COCO	35.9%	34	32	94.1%
MainSearch	23.8%	799	729	91.2%

structure while injecting necessary shortcuts to navigate the global structure.

Shortcut Utility and Trap Escape Rate. To validate that surviving shortcuts (Table 2) provide critical value rather than structural bloat, we profile their online search utility (Table 3). These explicit shortcuts are highly active, comprising 23% to 44% of search hops. Importantly, they rescue up to 95% of trapped queries that fail in base HNSW, confirming that joint pruning successfully retains topologically vital edges to overcome OOD blindness.

Manifold-Adaptive Memory Footprint. As shown by the bars in Figure 5, CAMP-HNSW acts as a manifold-adaptive rewirer rather than a uniform densifier. On the highly fragmented LAION manifold, our Joint Heuristic Pruning effectively identifies and removes redundant local edges, achieving a significantly tighter footprint—up to a 15.5% lower out-degree—compared to the additive NGFix. Conversely, on relatively clustered datasets like ImageNet,

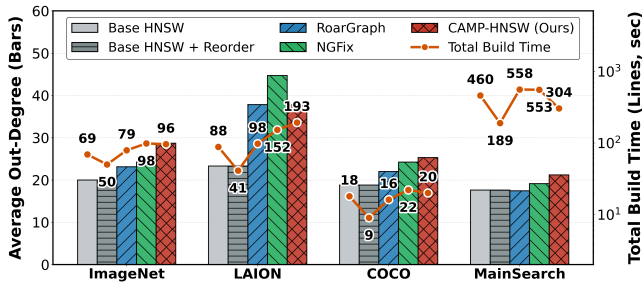


Figure 5: Comparison of average out-degree (bars, left-axis) and total offline build time (line, right-axis) across datasets.

COCO, and MainSearch, CAMP-HNSW dynamically constructs additional routing bridges to escape deep topological traps. While this results in a marginally higher average out-degree compared to the baselines, this structural expansion is strictly adaptive. Crucially, as we demonstrate in our ablation study (Section 4.5), simply inflating the graph with more edges significantly degrades routing throughput. Therefore, the superior QPS of CAMP-HNSW is not a trivial byproduct of an increased memory budget, but the direct result of highly targeted cross-manifold bridging.

Scalability and Hardware-Efficient Construction. The construction efficiency (lines in Figure 5) further highlights the scalability of CAMP-HNSW. On benchmarks like COCO and ImageNet, our system achieves build times strictly faster than NGFix. This demonstrates that the offline acceleration gained from our physical cache alignment effectively overtakes the computational overhead of the two-stage mining. While the high fragmentation of the LAION manifold requires more intensive bridging efforts, the architectural scalability of our system truly shines on large-scale datasets. On the MainSearch dataset, CAMP-HNSW finishes the full build in just 304 seconds—over 1.8× faster than both RoarGraph and NGFix—securing the second-fastest position immediately following the base HNSW.

Peak Memory Profiling. Although offline construction incurs higher transient peak memory due to prototyping overheads (e.g., Python-C++ bindings and parallel buffering), the final online static footprint remains strictly bounded (Figure 5), ensuring highly scalable deployment.

This significant gain in efficiency is the direct result of our Cache-Aligned Memory Reordering; by optimizing physical data locality prior to the heavy mining phase, we drastically reduce cache-miss penalties during the neighbor evaluations. This proves that CAMP-HNSW is optimized for industrial-scale deployments where construction throughput and search performance are equally important.

4.4 Hardware-Level Microbenchmarks

To isolate the hardware-level impact of our Cache-Aligned Memory Reordering, we profile the CPU performance counters using the Linux perf tool. Since this data reordering operates strictly as a physical memory booster without altering the logical graph routing, the total number of distance computations remains exactly

Table 4: Impact of Cache-Aligned Memory Reordering on L3 Cache and CPU Time.

Dataset	Δ LLC-Loads	Δ LLC-Misses	Δ CPU Time
ImageNet	↓ 69.9M (5.5%)	↓ 52.5M (4.4%)	↓ 754ms (6.0%)
LAION	↓ 46.9M (3.2%)	↓ 9.1M (0.7%)	↓ 276ms (2.0%)
COCO	↓ 142.3M (9.8%)	↓ 104.2M (8.0%)	↓ 1,153ms (9.6%)
MainSearch	↓ 206.5M (3.1%)	↓ 453.5M (7.7%)	↓ 6,282ms (7.5%)

Note: ↓ denotes absolute decrease (relative improvement %) vs. un-reordered baseline.

constant. However, as shown in Table 4, this physical alignment yields substantial system-level efficiency.

L3 Miss Reduction and Orthogonal Acceleration. By grouping topologically adjacent vectors, the majority of memory fetches during dense local routing are successfully absorbed by the L1 and L2 caches. As detailed in Table 4, this spatial locality considerably reduces L3 cache requests (LLC-Loads) and successfully eliminates over 453.5 million cache misses on the MainSearch dataset. Because this physical optimization is decoupled from the logical topology, it provides an orthogonal acceleration, translating to a 6.0% to 9.6% reduction in total CPU execution time on relatively clustered manifolds. On the highly fragmented LAION dataset, where routing relies heavily on bypassing local neighborhoods via OOD shortcuts, the acceleration is naturally moderated (2.0%). Nevertheless, CAMP-HNSW consistently extracts hidden hardware performance margins across environments.

Mitigating Cross-Manifold Penalties via Bounded Degree. As analyzed in Section 2.1, bridging OOD traps requires OOD shortcuts that overcome topological blindness. While these inter-manifold shortcuts may not always span large Euclidean distances, they inherently connect points across fundamentally different data distributions. Because our Phase 1 reordering physically clusters data by local manifold proximity, traversing such a cross-manifold shortcut inevitably disrupts spatial locality, which typically leads to L3 cache thrashing (or “cache pollution”).

However, Table 4 confirms that CAMP-HNSW successfully suppresses this hardware penalty. Even on the LAION dataset—which requires frequent cross-manifold jumps due to its severe fragmentation—the system still secures a net reduction in both LLC-Misses (9.1M) and overall CPU Time (276ms). This resilience is directly attributed to our Joint Heuristic Pruning. By forcing cross-manifold shortcuts and base edges to compete, Phase 3 strictly bounds the maximum local out-degree. Consequently, while an inter-manifold jump inevitably incurs an initial cache miss, the system limits cache pollution; it evaluates a constrained, highly optimized set of neighbors rather than flooding the cache with unbounded additive edges. This ensures that the reordered local clusters (Phase 1) remain efficiently staged in the cache hierarchy throughout the routing process.

4.5 Incremental Component Ablation

To rigorously validate the necessity of each architectural component in CAMP-HNSW, we conduct an incremental ablation study. We isolate the impact of our proposed modules by sequentially integrating them into the base HNSW index. Table 5 summarizes the system metrics across all four datasets. We evaluate the memory

Table 5: Incremental Ablation Study. ‘Fail’ indicates the target recall was unreachable. Best results are in bold; second-best are underlined.

Variant	Deg (↓)	T (s) (↓)	QPS@95 (↑)	QPS@99 (↑)
<i>ImageNet</i>				
1. Base HNSW	20.0	<u>69</u>	1,407	<i>Fail</i>
2. + Naive Injection	54.7	353	289	<i>Fail</i>
3. + Cand Pre-Filter	29.5	118	3,859	<u>1,613</u>
4. + Competitive Prune	<u>28.4</u>	119	4,049	1,501
5. Reorder Only	20.0	50	1,591	<i>Fail</i>
6. CAMP-HNSW (Ours)	28.7	96	<u>4,046</u>	1,626
<i>LAION</i>				
1. Base HNSW	23.3	<u>88</u>	1,096	<i>Fail</i>
2. + Naive Injection	63.9	1,331	<i>Fail</i>	<i>Fail</i>
3. + Cand Pre-Filter	43.5	257	2,221	744
4. + Competitive Prune	<u>37.7</u>	254	<u>2,536</u>	<u>831</u>
5. Reorder Only	23.3	41	1,016	<i>Fail</i>
6. CAMP-HNSW (Ours)	37.8	193	2,706	876
<i>COCO</i>				
1. Base HNSW	18.8	<u>18</u>	2,490	798
2. + Naive Injection	51.5	84	513	265
3. + Cand Pre-Filter	26.6	30	3,848	2,505
4. + Competitive Prune	25.6	30	4,019	<u>2,558</u>
5. Reorder Only	18.8	9	2,635	865
6. CAMP-HNSW (Ours)	<u>25.3</u>	20	4,093	2,616
<i>MainSearch</i>				
1. Base HNSW	17.6	460	1,561	<i>Fail</i>
2. + Naive Injection	40.1	1,518	263	<i>Fail</i>
3. + Cand Pre-Filter	21.7	571	6,857	2,115
4. + Competitive Prune	21.4	573	<u>7,024</u>	1,993
5. Reorder Only	17.6	189	1,489	<i>Fail</i>
6. CAMP-HNSW (Ours)	<u>21.2</u>	<u>304</u>	7,626	<u>2,021</u>

footprint (average out-degree, Deg), offline construction overhead (total build time in seconds, T), and online routing efficiency at critical precision targets (QPS@95 and QPS@99). The ‘Fail’ label indicates that the target recall threshold is unreachable by the graph within the query hop budget, signifying a catastrophic loss of navigational connectivity.

To isolate the search performance from transient system-level interference (e.g., OS background tasks) that inherently degrades throughput, all configurations were evaluated five independent times, and the median QPS is reported.

The Failure of Naive Densification. Naturally, the unmodified base index (Variant 1, Base HNSW) records the lowest average degree and the shortest build time, as it lacks any OOD adaptation mechanisms. However, without topological correction, it systematically fails to reach extreme recall targets (QPS@99). A straightforward attempt to resolve this via unconditional shortcut injection (Variant 2, + Naive Injection) severely degrades performance. By bypassing heuristic regulation, the out-degree approaches its physical limit (e.g., 23.3 $\rightarrow$ 63.9 on LAION), increasing the offline build time by 15 $\times$ and completely collapsing the operational throughput.

Algorithmic Efficiency via Two-Stage Pruning. Our two-stage pruning resolves this bottleneck. While the initial candidate

pre-filtering (Variant 3, + Cand Pre-Filter) acts as a robust baseline filter, the joint competitive pruning stage (Variant 4, + Competitive Prune) dictates the graph’s fundamental algorithmic efficiency. By actively forcing original base edges and new shortcuts to compete, Variant 4 significantly slims down the topology compared to naive densification. This structural rewiring ensures that redundant local edges are seamlessly replaced by critical OOD shortcuts, translating a refined topology directly into significant QPS improvements at the 95% recall target.

Hardware-Level Acceleration & Cold-Start Adaptability.

To isolate the hardware impact of Phase 1 and demonstrate baseline acceleration in a strict 0% cold-start scenario (no historical queries), we introduce Variant 5 (Reorder Only). By applying Cache-Aligned Memory Reordering directly to the Base HNSW, Variant 5 (Table 5) provides immediate benefits without queries: it significantly reduces offline build time (e.g., 69s $\rightarrow$ 50s on ImageNet) and yields “free” throughput gains for basic routing (e.g., QPS@95 improves from 1,407 to 1,591). Crucially, Variant 5 systematically fails to reach the 99% recall target across all datasets. This confirms that while memory layout optimization provides a baseline throughput boost, the capacity to overcome topological blindness and reach extreme precision is strictly driven by the OOD-aware rewiring of Phases 2 and 3.

The system-level effectiveness of our complete co-design is ultimately demonstrated by Variant 6 (CAMP-HNSW (Ours)), which integrates this physical memory alignment with the pruned topology of Variant 4. A profound observation from Table 5 is that adding this physical reordering step actually reduces the overall build time compared to Variant 4 (e.g., 119s $\rightarrow$ 96s on ImageNet). Counter-intuitively, on the MainSearch dataset, Variant 6 finishes construction in 304s—significantly faster than the unmodified base HNSW (460s). Because the dataset is physically reordered prior to any graph construction, it drastically minimizes the L3 cache miss penalties common in large-scale neighborhood evaluations during the building and mining phases.

Crucially, this offline acceleration translates into a direct online advantage. The physically clustered memory synergizes perfectly with our pruned topology, securing the peak QPS at the most strict precision target (QPS@99%) across ImageNet, LAION, and COCO. On the MainSearch dataset, a minor trade-off emerges: Variant 3 retains a slightly higher degree, granting marginal benefits at the extreme 99% boundary. However, our full co-design (Variant 6) purposefully trims this excess degree to secure a substantial throughput increase at the 95% recall target (reaching 7,626 QPS) while nearly halving the offline build time. This confirms that CAMP-HNSW achieves an optimal system balance: drastically accelerating the raw construction speed on large-scale manifolds while delivering highly efficient routing throughput.

4.6 Default Parameter Sensitivity

A common critique of robust graph indices is their heavy reliance on exhaustive, dataset-specific hyperparameter tuning. To demonstrate that CAMP-HNSW operates efficiently with fixed default parameters, we analyze the parameter sensitivity of our primary offline components. Crucially, to prevent overfitting, we derive our parameters solely using the 1M-scale datasets (ImageNet and

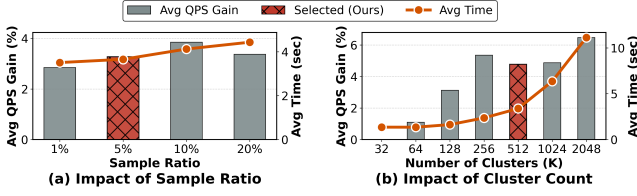


Figure 6: Parameter sensitivity analysis for Cache-Aligned Memory Reordering evaluating the impact of sample ratio and cluster count on ImageNet and LAION.

LAION) as reference benchmarks. We then show that this fixed configuration generalizes directly to both large-scale (MainSearch, 11.2M) and high-dimensional (COCO, 768) environments without further tuning.

Memory Reordering: Hardware-Level Granularity. The Cache-Aligned Memory Reordering is governed by two parameters: the dataset sampling ratio (Sam) for training centroids, and the total number of spatial clusters (K). Empirical evaluations on the 1M reference datasets (Figure 6) indicate that while increasing the sample ratio (e.g., 10%) and cluster count (e.g., $K = 2048$) maximizes the QPS gain, it incurs a substantial computational penalty. Furthermore, unconditionally adopting such empirical peaks at large-scale production scales (e.g., $\geq 10M$) introduces prohibitive preprocessing bottlenecks. To ensure scalable and stable operation, CAMP-HNSW derives a deterministic configuration ($Sam = 5\%$, $K = 512$).

Regarding the sample ratio, while a 10% sample incurs only a marginal offline time increase at the 1M scale, this computational overhead scales linearly with the dataset size and becomes a primary bottleneck on large-scale datasets like MainSearch. The 5% ratio acts as a statistical sweet spot, effectively capturing the macroscopic manifold structure while halving the data footprint for centroid training. Regarding the cluster count, K is governed by our deterministic L3 cache rule (Section 3.3). For MainSearch ($N = 11.2M$, $D = 256$), the theoretical lower bound to safely fit within our 33.75 MB L3 cache is $K \geq 340.1$. Setting $K = 256$ would create ≈ 44.5 MB memory chunks, increasing cache misses during routing. Thus, $K = 512$ provides a safe operational boundary. By trading off a marginal empirical QPS gain on the reference datasets, this formula preserves hardware efficiency across extreme environments without heuristic tuning.

Our L3 cache formula serves as a principled heuristic to optimize spatial locality for neighborhood traversals. Although routing does not load entire clusters simultaneously, this physical alignment maximizes the probability of staging topologically adjacent candidates into adjacent cache lines, significantly mitigating random memory access penalties.

Shortcut Mining: Pareto Frontier and Knee Point. The selection of mining parameters—search depth (ef_{mine}) and maximum shortcut candidates (N_{mine})—dictates the trade-off between offline construction cost and online precision. We conducted an extensive grid search on the reference datasets, varying $ef_{mine} \in [10, 500]$ and $N_{mine} \in [10, 100]$. Figure 7 visualizes the resulting Pareto frontier. The curve demonstrates a clear pattern of diminishing

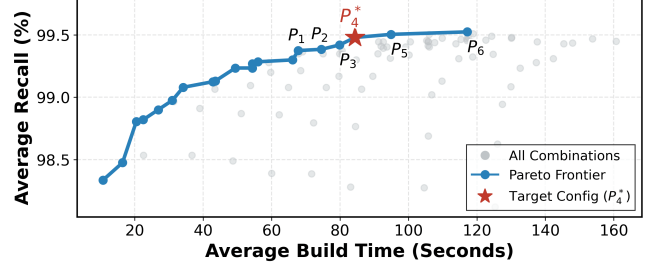


Figure 7: Global Pareto frontier of average build time vs. average recall across ImageNet and LAION. The optimal configuration P_4^* and neighboring states are denoted as (ef_{mine}, N_{mine}) : $P_1(30, 60)$, $P_2(150, 60)$, $P_3(100, 70)$, $P_4^*(150, 70)$, $P_5(150, 80)$, $P_6(150, 100)$.

marginal returns. Modest initial investments yield steep precision gains, confirming the necessity of OOD exploration. However, the curve sharply flattens after the empirical “knee” point at $(ef_{mine} = 150, N_{mine} = 70)$.

Pushing the parameters beyond this threshold (e.g., to $N_{mine} \geq 80$) incurs a notable build time penalty due to the $O(N_{mine}^2)$ complexity of cross-manifold candidate generation. Furthermore, these excessive candidates provide negligible recall gains ($\leq 0.02\%$), as our Joint Heuristic Pruning effectively filters out spatially redundant edges anyway. Therefore, we explicitly select $(ef_{mine} = 150, N_{mine} = 70)$ as the optimal and most cost-effective operational boundary.

Robustness and Reproducibility. By combining robust topological pruning with this hardware-aware clustering setup ($ef_{mine} = 150$, $N_{mine} = 70$, $Sam = 5\%$, $K = 512$), CAMP-HNSW eliminates the burden of per-dataset tuning. For full reproducibility, we explicitly fixed the remaining MiniBatchKMeans parameters to lightweight constants ($batch_size = 2048$, $max_iter = 50$, $n_init = 1$) as they showed marginal impact. It is exactly this single, fixed parameter combination that drives the dominant Pareto frontiers and superior routing efficiency demonstrated earlier in Section 4.2, proving that CAMP-HNSW delivers robustly superior performance across diverse data manifolds without relying on dataset-specific overfitting.

4.7 Extreme Robustness: ID and Cold-Start

A truly practical ANNS must not only handle OOD shifts but also maintain high efficiency under extreme operational constraints. We evaluate CAMP-HNSW under two such conditions: 1) standard ID workloads, where OOD adaptations typically cause unnecessary overhead, and 2) “cold-start” scenarios, where historical calibration queries are highly sparse.

Zero Routing Penalty on ID Workloads. Densifying a graph to bridge OOD gaps often disrupts optimal routing paths for simple ID queries, resulting in degraded throughput. To verify that CAMP-HNSW avoids this trade-off, we evaluate the models on GloVe and the ID-variant of ImageNet. As shown in Figure 8 (left), on the highly uniform GloVe dataset, all models exhibit nearly identical QPS trajectories. This proves that our Joint Heuristic Pruning successfully recognizes the topological simplicity of the ID manifold and suppresses redundant shortcut generation, preventing any

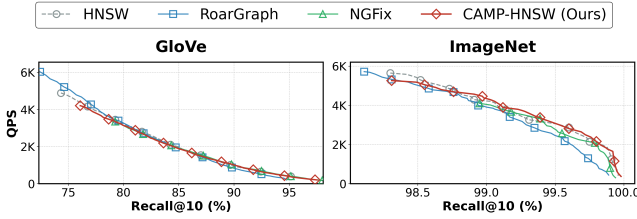


Figure 8: Performance comparison of CAMP-HNSW and baseline algorithms on ID datasets.

throughput degradation. Furthermore, on ImageNet (ID) (Figure 8, right), CAMP-HNSW consistently surpasses all baselines at the $\geq 99\%$ recall regime. This acceleration is the direct result of our Cache-Aligned Memory Reordering, which makes distance computations highly hardware-efficient even for standard workloads.

Data Efficiency and Cold-Start Resilience. A critical challenge for query-aware graphs is their heavy reliance on massive historical query logs. In practical “cold-start” scenarios, accumulating such massive logs is challenging, making data efficiency a critical operational requirement. Figure 9 illustrates the achieved QPS at the extreme $\geq 99\%$ recall boundary when varying the available training queries from 10% to 100%.

Baseline methods exhibit a critical vulnerability to data sparsity, essentially overfitting to large calibration sets. The base HNSW, RoarGraph, and NGFix all suffer severe performance drops or fail completely to reach the 99% recall target when restricted to $\leq 25\%$ training data. In stark contrast, CAMP-HNSW demonstrates extreme sample efficiency. Under the most severe cold-start condition (10% ratio), it delivers up to $2.9\times$ higher QPS than RoarGraph (on COCO) and $1.5\times$ higher QPS than NGFix (on LAION). Impressively, by utilizing just 25% of the queries, CAMP-HNSW fully surpasses the peak performance of RoarGraph operating on 100% of the data. This pronounced performance gap is robustly sustained across all datasets and query ratio intervals.

Ultimately, unlike baseline methods that heavily rely on memorizing historical routing paths for additive edge injection, our Joint Heuristic Pruning analyzes and targets the intrinsic topological bottlenecks of the manifold. This allows CAMP-HNSW to accurately identify critical OOD gaps even from heavily subsampled query logs, while simultaneously preventing structural bloat on simple ID manifolds. These results definitively establish CAMP-HNSW as a highly robust and hardware-efficient in-memory index.

Furthermore, CAMP-HNSW supports step-wise adaptation. At a 0% cold-start, the system deploys only Phase 1 for immediate acceleration (Variant 5 in Table 5). With minimal logs (e.g., 10%), it incrementally executes Phases 2 and 3 to fully robustify the topology against OOD gaps.

Robustness to Noisy Labels. CAMP-HNSW also exhibits extreme systemic robustness when exact calibration GTs are unavailable. Even with 90% purely random noisy labels, it strictly outperforms the base graph. Phase 3 acts as a structural regularizer, dynamically discarding non-monotonic noisy edges in favor of valid shortcuts or true local connections.

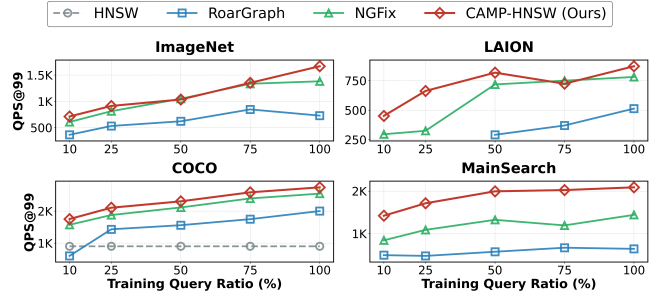


Figure 9: Data efficiency evaluation: QPS@99 across varying calibration query ratios (10% ~ 100%).

5 RELATED WORK

Proximity graphs, such as HNSW [36], NSG [13], and Vamana [24], are the *de facto* standard for high-performance ANNS. However, their structural integrity relies on local isotropy, making greedy routing fragile when exposed to severe distribution shifts (e.g., cross-modal retrieval or LLM attention mechanisms [33]). To mitigate this, recent query-aware frameworks aim to explicitly bridge inter-manifold gaps. For instance, OOD-DiskANN [23] incorporates OOD queries for disk-based routing, RoarGraph [6] projects historical queries into a bipartite structure, and NGFix [20] dynamically injects supplementary routing edges. However, as analyzed in Section 2.2, these additive densification strategies lack holistic edge re-evaluation, thereby accumulating redundant connections that inflate the memory footprint and degrade hardware cache efficiency during dense routing.

Furthermore, as datasets scale, ANNS bottlenecks frequently shift to memory bandwidth and cache latency [7, 30, 54]. While studies have explored clustering-based layouts to accelerate ID search [47, 51], optimizing the physical memory layout for dynamic, OOD-augmented proximity graphs remains challenging. Finally, while our prior works (DHNSW [26] and PRO-HNSW [27]) successfully addressed ID search adaptability and proactive index repair, CAMP-HNSW targets a fundamentally orthogonal dimension. By integrating Cache-Aligned Memory Reordering with Joint Heuristic Pruning, our system uniquely resolves both cross-modal topological blindness and hardware-level bottlenecks within a rigorously bounded memory footprint.

6 CONCLUSION

We proposed CAMP-HNSW, a Pareto-superior framework that defends against OOD shifts by synergizing targeted topological rewiring with cache-aligned physical memory layouts. Extensive evaluations demonstrate SOTA throughput across 95%-99% recall using default parameters. By reducing memory footprints by 15.5% on fragmented manifolds, CAMP-HNSW provides a scalable, hardware-efficient solution for multi-modal databases.

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